

The Data Trust Index: A Multidimensional Framework for Evaluating Health Data Integrity in AI Systems

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Abstract

Health data powering artificial intelligence systems varies widely in provenance, consent coverage, temporal currency, and clinical validation, yet most AI pipelines treat data as uniformly trustworthy once ingested. This paper introduces the Data Trust Index (DTI), a multidimensional scoring framework that assigns a continuous integrity score from 0 to 100 to health data records prior to their use in downstream AI inference. DTI decomposes trust across eight weighted dimensions: Provenance, Consent, Recency, Quality, Concordance, Validation, Breadth, and Stability. We describe the theoretical basis for each dimension, the time-decay functions applied to temporal signals, and the concordance methodology used to corroborate signals across independent sources. We present a production implementation case study with imaware, a direct-to-consumer diagnostics company operating 105,000 diagnostic records, where DTI-based segmentation reduced data preparation cycles from three weeks to two hours and surfaced a customer segment accounting for 20% of revenue that was invisible in fragmented data views. We situate DTI within the regulatory landscape of the 21st Century Cures Act, the FDA's AI/ML-Based Software as a Medical Device (SaMD) guidance, and TEFCA, arguing that a standardized trust metric for health data is a prerequisite for safely deploying clinical AI at scale. The DTI framework and its underlying ALDR engine are covered by U.S. patent application SuperTruth0010CP1.

Keywords: health data quality, data integrity, AI/ML in healthcare, data provenance, consent management, clinical decision support

ACM CCS: Computing methodologies → Machine learning → Machine learning approaches; Applied computing → Life and medical sciences → Health informatics; Security and privacy → Privacy protections → Data anonymization and sanitization

1. Introduction

The dominant assumption in machine learning pipeline design is that data, once collected and stored, can be treated as fixed input. In health AI, this assumption is dangerous. A fasting glucose measurement from a CLIA-certified laboratory with documented chain of custody carries fundamentally different epistemic weight than a self-reported blood sugar entry in a consumer wellness app. Yet most clinical AI systems ingest both without distinction, relying on dataset-level quality audits rather than record-level trust assessment [1, 2].

The consequences are not hypothetical. The FDA’s 2021 action plan for AI/ML-Based Software as a Medical Device explicitly identified “good ML practices” as a gap area, citing the absence of standardized methods for characterizing training and inference data quality [3]. Research in clinical NLP has documented that downstream model performance degrades measurably when training data contains unlabeled missingness patterns, consent boundary violations, or temporal staleness [4, 5]. In federated learning environments, where data never leaves its source institution, the absence of a common quality signal makes model aggregation across sites an act of statistical faith [6].

The problem is structural. Health data originates across dozens of modalities (laboratory assays, wearable biometrics, clinical notes, claims, genomics, SDOH surveys) with incompatible provenance standards, variable consent frameworks, and modality-specific temporal decay rates. No existing standard provides a unified, record-level signal that an AI system can condition on before incorporating a data point into training or inference.

This paper introduces the Data Trust Index (DTI), a framework that produces exactly such a signal. DTI scores each health data record on a 0-100 scale by evaluating eight dimensions of data quality and trustworthiness. The score is composable (dimensions can be reweighted for specific use cases), interpretable (each dimension contribution is reported separately), and actionable (score tiers map to recommended use contexts from exploratory research to regulatory submission).

The remainder of this paper is structured as follows. Section 2 reviews related work in health data quality, data provenance, and trust metrics. Section 3 defines the eight DTI dimensions and their default weights. Section 4 describes the scoring methodology, including time-decay functions and concordance calculations. Section 5 describes the SuperTruth pipeline implementing DTI at scale. Section 6 presents the imaware case study. Section 7 discusses regulatory implications. Section 8 concludes.

2. Background and Related Work

2.1 Health Data Quality Frameworks

Health data quality has been studied extensively in the context of electronic health records (EHRs). Weiskopf and Weng [7] proposed a framework for EHR data quality that identified completeness, correctness, concordance, plausibility, and currency as key dimensions. The ONC’s Interoperability Standards Advisory and HL7 FHIR quality profiles address structural conformance but not epistemic trustworthiness [8]. The RECORD statement for observational studies and the EQUATOR network’s reporting guidelines address how data should be described in publications but do not provide a runtime scoring mechanism [9].

In the clinical trials domain, ICH E6(R2) Good Clinical Practice guidelines and FDA 21 CFR Part 11 address data integrity for regulatory submissions, focusing on audit trails, electronic signatures, and system validation [10]. These frameworks are rigorous but narrow: they apply to controlled trial environments, not the heterogeneous data streams feeding real-world AI systems.

2.2 Data Provenance

Provenance research in computer science has produced the PROV-DM standard [11], W3C's provenance model for linked data, and domain-specific extensions for scientific workflows [12]. In healthcare, the HL7 FHIR Provenance resource provides a schema for recording data origin and transformation history, but adoption is inconsistent and the resource does not encode a quality signal [13]. Buneman et al.'s foundational work on "why provenance" [14] distinguishes between why a data item appears in a result and where it came from, a distinction directly relevant to health data where source pedigree (CLIA-certified lab vs. consumer device) matters independently of chain-of-custody completeness.

2.3 Consent and Data Governance

The HIPAA Privacy Rule [15] and the HITECH Act [16] establish minimum consent requirements for protected health information, but neither specifies how consent scope should be encoded in data systems or how consent completeness should be scored. The 21st Century Cures Act [17] introduced information blocking provisions and mandated TEFCA as a national interoperability framework, but TEFCA's trust framework addresses organizational participation, not individual record-level consent quality [18]. GDPR Article 7 [19] requires demonstrable consent but provides no scoring mechanism. The Common Rule [20] governs research consent but does not generalize to clinical or commercial contexts.

2.4 Temporal Data Quality

Temporal aspects of data quality have received attention in database research [21] and in clinical informatics. Xiao et al. [22] demonstrated that EHR-based models degrade when trained on features with high temporal missingness. In wearable data specifically, Bent et al. [23] showed that heart rate variability features have meaningfully different stationarity properties over time than discrete biomarker panels, motivating modality-specific decay modeling.

2.5 Gap

The frameworks reviewed share a structural limitation: they are designed for documentation, reporting, or static audit, not for runtime use. Weiskopf and Weng's five-dimension framework has been cited over 800 times and represents the dominant approach to EHR data quality characterization, but it produces no score and provides no mechanism for conditioning a machine learning pipeline on data quality at inference time. The RECORD statement and ICH E6 guidelines are publication and trial-design standards; they cannot be applied to individual records flowing through a production system. FDA's SaMD action plan identifies training data quality as a critical gap but defines no quantitative standard for addressing it.

The consent gap is particularly acute. HIPAA and GDPR establish legal requirements for consent but specify neither how to encode consent completeness in a data record nor how to propagate consent revocation to downstream consumers. No existing framework provides a runtime, per-record signal that an AI system can use to confirm that the data it is incorporating has valid, current, appropriately scoped consent.

The temporal gap is similarly unaddressed. All existing frameworks treat data quality as a static property measured at a point in time. In production AI systems that re-use historical data for ongoing inference, a record’s quality degrades as it ages. No existing framework applies modality-specific decay functions to produce a recency-adjusted score that reflects a record’s current trustworthiness rather than its trustworthiness at the time of collection.

DTI addresses all three gaps simultaneously: it produces a runtime, per-record score that integrates provenance, consent, temporal currency, quality, cross-source concordance, clinical validation, dimensional richness, and signal stability into a single auditable signal usable in production AI pipelines.

3. The DTI Framework

DTI models trustworthiness as a weighted sum of eight dimension scores, each normalized to 0-100. The default weight vector reflects the relative evidentiary importance of each dimension for clinical AI applications, determined through a combination of regulatory guidance review, clinical expert elicitation, and empirical analysis of feature sensitivity in downstream model performance.

3.1 Dimension Definitions

Table 1: DTI Dimensions, Weights, and Scoring Anchors

#	Dimension	Weight	Low Score (0-40)	High Score (80-100)
1	Provenance	25%	Unknown source, no chain of custody	CLIA/CAP-certified lab, full chain of custody documented
2	Consent	20%	Implied or absent consent, no revocation mechanism	Explicit, scoped, time-bounded consent with documented revocation path
3	Recency	15%	Stale beyond modality-specific threshold	Collected within decay floor for the modality
4	Quality	10%	High missingness, noise above clinical threshold	Complete fields, noise below floor, no imputation flags
5	Concordance	10%	Single source, no corroboration	Corroborated across two or more independent sources
6	Validation	10%	No clinical linkage, no outcome correlation	Peer-reviewed evidence linking biomarker to clinical outcome

#	Dimension	Weight	Low Score (0-40)	High Score (80-100)
7	Breadth	5%	Single modality, minimal contextual signals	Multi-modal coverage: biomarkers, biometrics, contextual/SDOH
8	Stability	5%	High intra-individual variance, poor test-retest	Low variance, strong test-retest reliability documented

Total: 100%

3.2 Dimension Details

Provenance (25%). Provenance encodes source pedigree and custody chain. The highest scores require CLIA (Clinical Laboratory Improvement Amendments) or CAP (College of American Pathologists) certification for laboratory data, documented instrument calibration, and an unbroken audit trail from collection to storage. Consumer-generated data (wearable sensors, self-report) receives provenance scores conditioned on device FDA clearance status and documented calibration protocols. Each custody transfer in the chain is scored for documentation completeness; gaps propagate downward.

Consent (20%). Consent scoring evaluates four sub-dimensions: explicitness (implied vs. informed vs. explicit), scope (how narrowly the consent defines permitted uses), duration (whether the consent includes an expiration or requires renewal), and revocation hygiene (whether a documented mechanism exists and has been tested). Consent tiers map to five levels in SuperTruth’s ConsentOS layer, from implicit collection through regulatory-grade consent with cryptographic attestation.

Recency (15%). Recency applies modality-specific time-decay functions rather than a uniform staleness threshold. A complete blood count (CBC) ages more rapidly than a stable genomic variant; continuous HRV data from a wearable decays differently than an annual lipid panel. Each modality in the DTI taxonomy has a decay half-life derived from clinical literature on within-individual stability. The recency score at time t for a measurement taken at time t_0 is computed as a function of elapsed time and modality-specific decay parameters, described in Section 4.

Quality (10%). Quality captures structural completeness, missingness rate, and signal-to-noise characteristics. Missingness is scored at the field level; records with imputed values receive partial credit conditional on the imputation method documented. Noise floor assessment uses modality-specific clinical reference ranges: a value three standard deviations outside the reference range without a documented clinical explanation triggers a quality flag.

Concordance (10%). Concordance scores the degree to which a measurement is corroborated by independent sources measuring the same construct. A self-reported weight that matches a scale measurement from a clinical visit within 30 days receives high concordance. A laboratory glucose value that conflicts with a contemporaneous CGM reading without documented explanation receives low concordance. Independence of sources is assessed structurally: two measurements from the same device or the same EHR system are not treated as independent.

Validation (10%). Validation scores the clinical evidence base for the biomarker or measurement type. Biomarkers with strong peer-reviewed evidence linking them to clinical outcomes (e.g., HbA1c and diabetes outcomes, LDL-C and cardiovascular

events) receive high validation scores. Novel or exploratory biomarkers with limited evidence base receive lower scores. This dimension is a property of the measurement type, not the individual record, and is updated as the literature evolves.

Breadth (5%). Breadth scores dimensional richness: the number of distinct modalities represented for a given patient or record set. A record set that includes laboratory panels, biometric wearable data, clinical notes, and SDOH signals receives higher breadth scores than a set containing only claims data. Breadth penalizes over-reliance on any single data modality.

Stability (5%). Stability scores within-individual measurement variance and test-retest reliability. High stability scores require documented low intra-individual coefficient of variation (CV) for the measurement type and, where available, direct test-retest data. Stability is particularly important for biomarkers used as monitoring signals in longitudinal AI applications.

3.3 Score Tiers

DTI scores map to four operational tiers:

- **Bronze (55-69):** Suitable for exploratory analysis, hypothesis generation, and non-clinical research.
- **Silver (70-79):** Suitable for operational analytics, population health, and commercial AI applications.
- **Gold (80-89):** Suitable for clinical decision support and patient-facing applications.
- **Platinum (90+):** Suitable for regulatory submission, clinical trial support, and high-stakes AI inference.

3.4 Weight Justification and Sensitivity

The default weight vector was developed through a three-phase process. First, a regulatory guidance review mapped each dimension to requirements in HIPAA, 21st Century Cures, FDA SaMD guidance, and ICH E6(R2), identifying which dimensions are most frequently cited as failure modes in regulatory enforcement actions and adverse event reports. Second, a structured elicitation with clinical informaticists and regulatory specialists rank-ordered dimensions by their estimated contribution to downstream model failure in clinical AI contexts. Third, a preliminary sensitivity analysis examined how composite score orderings responded to single-dimension weight perturbations of ± 5 percentage points. Across the dimensions examined, score orderings showed high stability under moderate weight variation. Prospective, controlled sensitivity studies across multiple deployment contexts are included in the future work agenda (Section 7.4).

Provenance receives the highest weight (25%) because source pedigree errors are the least recoverable class of data quality failure: a record from an unknown source cannot be remediated by downstream processing. Consent receives the second-highest weight (20%) because consent violations carry direct regulatory and legal consequences independent of data quality. Recency (15%) ranks third because temporal staleness is the most common cause of silent model degradation in production clinical AI deployments.

Institutions with domain-specific requirements may rebalance weights within the constraint that no single dimension exceeds 40% of the composite, preserving multi-dimensional coverage.

3.5 Comparison to Existing Frameworks

Table 2: DTI Compared to Existing Health Data Quality Frameworks

Framework	Dimensions	Record-Level Score	Runtime Applicable	Regulatory Mapping	Consent Scoring
Weiskopf & Weng (2013) [7]	5 (completeness, correctness, concordance, plausibility, currency)	No	No	Partial	No
ONC Interoperability Standards (FHIR/USCDI)	Structural conformance, completeness	No	No	ONC rules, information blocking	No
TEFCA Trust Framework	Organizational participation	No	No	ONC/TEFCA QHINs	No
FDA SaMD Action Plan (2021) [3]	Training data documentation	No	No	FDA	No
ICH E6(R2) GCP [10]	Trial data integrity	No	Trial environments only	FDA/EMA	Yes (narrow)
RECORD Statement [9]	Reporting completeness	No	No	None	No
DTI (this work)	8	Yes (0-100)	Yes	FDA, HIPAA, TEFCA, Cures Act	Yes (5 tiers)

The primary differentiator of DTI is its runtime applicability at the record level. Existing frameworks are designed for documentation, reporting, or static audit; none produces a score that can be used as an input feature or filter in a live AI pipeline.

4. Scoring Methodology

4.1 Composite Score

The DTI composite score S for record r is computed as:

$$S(r) = \sum_i w_i \cdot d_i(r)$$

where w_i is the weight for dimension i and $d_i(r)$ is the normalized dimension score (0-100) for record r on dimension i . Default weights are defined in Table 1. For domain-specific deployments (e.g., oncology trial data where regulatory-grade consent is

mandatory), weights can be rebalanced by the deploying organization within constraints defined in the SuperTruth configuration layer.

Algorithm 1: DTI Record Scoring

Input: record r , weight vector W , modality m , timestamp t_{collect}
Output: composite score $S(r)$, per-dimension scores $D(r)$, confidence interval $CI(r)$

1. $P \leftarrow \text{score_provenance}(r.\text{source}, r.\text{custody_chain}, r.\text{certification})$
2. $Co \leftarrow \text{score_consent}(r.\text{consent_type}, r.\text{consent_scope}, r.\text{consent_expiry}, r.\text{revocation_status})$
3. $\Delta t \leftarrow \text{current_time} - t_{\text{collect}}$
4. $Re \leftarrow \text{recency_decay}(\Delta t, m)$ // Section 4.2
5. $Q \leftarrow \text{score_quality}(r.\text{fields}, r.\text{missingness}, r.\text{noise_floor}, m)$
6. if $r.\text{has_independent_source}$ then
7. $C \leftarrow \text{concordance_score}(r, r.\text{independent_source}, m)$ // Section 4.3
8. else
9. $C \leftarrow 50$ // neutral default
10. $Va \leftarrow \text{lookup_validation_score}(m)$ // literature-derived, not record-specific
11. $Br \leftarrow \text{score_breadth}(r.\text{modalities})$
12. $St \leftarrow \text{score_stability}(r.\text{variance_history}, m)$
13. $D(r) \leftarrow [P, Co, Re, Q, C, Va, Br, St]$
14. $S(r) \leftarrow \sum_i W[i] \cdot D(r)[i]$
15. $CI(r) \leftarrow \text{propagate_confidence}(D(r), r.\text{documentation_completeness})$
16. return $S(r), D(r), CI(r)$

Lines 6-9 reflect the asymmetric treatment of concordance: the absence of an independent source is not penalized (defaulting to neutral 50) rather than scored negatively, because many legitimate data sources have no contemporaneous independent measurement available.

4.2 Recency Decay Functions

For each modality m with decay half-life τ_m (in days), the recency score at elapsed time Δt is:

$$R(\Delta t, m) = 100 \cdot \exp(-\lambda_m \cdot \Delta t)$$

where $\lambda_m = \ln(2) / \tau_m$. Scores are floored at 0. The decay function is applied from the timestamp of collection, not receipt, where collection timestamp is documented.

Table 2: Modality-Specific Recency Decay Half-Lives

Modality	Half-Life τ (days)	Clinical Basis
Complete blood count (CBC)	90	Erythrocyte and leukocyte turnover rates

Modality	Half-Life τ (days)	Clinical Basis
Lipid panel	180	Cardiovascular risk signal stability
HbA1c	90	Reflects 3-month erythrocyte lifecycle
Continuous HRV aggregates	14	High intra-individual variability; short stationarity window
Fasting glucose (discrete)	30	Metabolic state changes on weekly-monthly timescale
SDOH survey data	365	Social determinants change slowly relative to biomarkers
Clinical notes (structured fields)	180	Diagnosis and medication states are semi-stable
Genomic variants	∞	Constitutional variants do not change; scored on collection quality only
Consumer wearable daily aggregates	30	Device accuracy degrades; behavioral patterns shift

Half-life values are derived from clinical literature on within-individual measurement stability and represent the elapsed time at which the recency score falls to 50 (i.e., the measurement retains half its original recency credit). Organizations may calibrate half-lives to their specific clinical context.

4.3 Concordance Calculation

For a target measurement m from source s_1 , concordance with independent measurement m' from source s_2 at time proximity δt is:

$$C(m, m') = \max(0, 100 \cdot (1 - |m - m'| / \text{tolerance}(\text{type})))$$

where $\text{tolerance}(\text{type})$ is the clinically acceptable measurement variation for the measurement type. If no contemporaneous independent measurement exists, concordance defaults to 50 (neutral, not penalized). Concordance scores above 85 for a measurement pair elevate the composite score; scores below 30 trigger a concordance flag that is returned to the consuming application.

Example. A fasting glucose measurement of 110 mg/dL from an at-home collection kit is compared against a clinical laboratory value of 113 mg/dL drawn four days later. The clinically accepted measurement tolerance for fasting glucose from point-of-care devices is ± 15 mg/dL per ISO 15197:2013 [24] and FDA 21 CFR 862.1345. The concordance score is:

$$C = \max(0, 100 \cdot (1 - |110 - 113| / 15)) = \max(0, 100 \cdot (1 - 0.20)) = 80$$

This score (80) indicates good concordance. The two measurements agree within clinical tolerance despite originating from independent collection events and different laboratory systems. If the at-home value had been 130 mg/dL (a 20 mg/dL discrepancy exceeding

the 15 mg/dL tolerance), the concordance score would be $\max(0, 100 \cdot (1 - 1.33)) = 0$, triggering a concordance flag for clinical review.

4.4 Confidence Intervals and Propagation

Each dimension score includes a confidence interval derived from data completeness. When provenance documentation is partial, the provenance score carries a wider interval. The composite score’s confidence interval is propagated from dimension-level intervals using standard error propagation. Consuming applications can filter records by composite score confidence in addition to point estimate.

5. Implementation: The SuperTruth Pipeline

SuperTruth implements DTI through a four-stage pipeline illustrated in Figure 1.

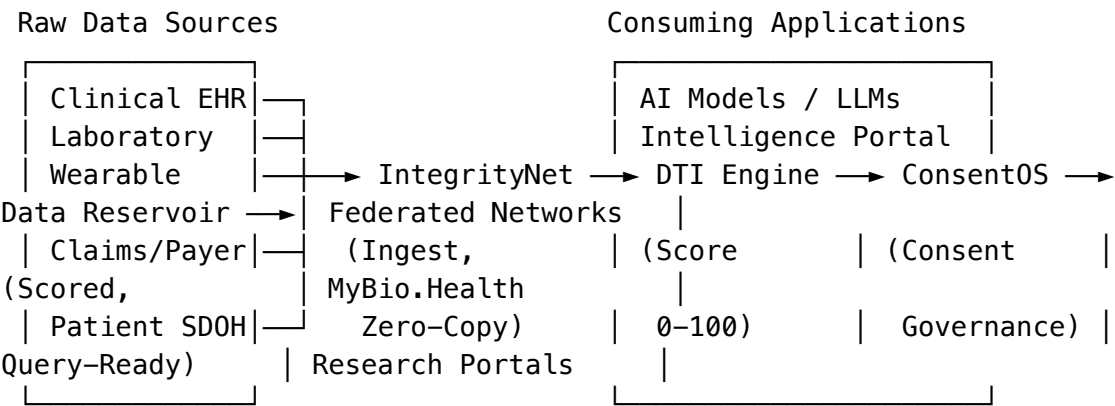


Figure 1: The SuperTruth four-stage pipeline. Data from heterogeneous sources enters IntegrityNet via zero-copy connectors. Each record is scored by the DTI Engine across eight dimensions. ConsentOS enforces consent boundaries. The Data Reservoir exposes scored, consent-governed data to consuming applications. DTI score travels permanently with every record through all stages.

Table 3: Pipeline Stage Summary

Stage	Input	Operation	Output	Patent Coverage
IntegrityNet	Raw records (any source, any format)	Schema normalization, provenance capture, audit trail creation	Structured records with provenance metadata	SuperTruth0010CP1
DTI Engine	Provenance-annotated records	Eight-dimension scoring via ALDR (Algorithm 1)	Records with DTI score + per-dimension breakdown + confidence interval	SuperTruth0010CP1

Stage	Input	Operation	Output	Patent Coverage
ConsentOS	Scored records	Consent state enforcement, cryptographic attestation, revocation propagation	Consent-governed records with use-authorization flags	SuperTruth0010CP1
Data Reservoir	Scored, consent-governed records	Score-indexed storage, query interface, tier-based access control	Queryable data layer with DTI as first-class filter	SuperTruth0010CP1

SuperTruth implements DTI through a four-stage pipeline, described below:

Stage 1: IntegrityNet (Ingestion). Data is ingested via zero-copy connectors to source systems: FHIR R4 APIs, HL7 v2 feeds, SFTP drops from laboratory systems, and direct API integrations with wearable platforms. Zero-copy means source data is never moved to a SuperTruth data store; instead, a metadata record describing the data and its provenance is created. This architecture eliminates a class of consent and security risk associated with data replication.

Stage 2: DTI Engine (Scoring). The DTI Engine evaluates each ingested record against the eight dimensions. The engine is implemented using the ALDR (Adaptive Learning Data Resilience) architecture, which comprises three components: MEA (Matrix Exponential Attention) for cross-source signal alignment, SADC (Self-Adjusting Decay Control) for modality-specific temporal scoring, and GLMN (Graph-Laplacian Memory Networks) for longitudinal stability assessment. ALDR is the subject of U.S. patent application SuperTruth0010CP1.

Stage 3: ConsentOS (Consent Governance). ConsentOS enforces consent boundaries on scored data through five additive patient consent actions: Grant, Modify, Pause, Resume, and Revoke. Each action is recorded with a cryptographic attestation. Revocation propagates to all downstream systems in real time, ensuring that a patient's withdrawal of consent is honored across the entire pipeline without manual intervention. ConsentOS maintains an immutable audit log of every consent state transition.

Stage 4: Data Reservoir (Query-Ready Output). Scored, consent-governed data is exposed to consuming applications through a query interface that includes DTI score as a first-class filter. Applications can query for all records above a DTI threshold, within a score tier, or filtered by individual dimension scores. The reservoir is read-only from the perspective of data consumers; writes flow only through IntegrityNet.

6. Retrospective Deployment Study: imaware

This section presents results from a retrospective before/after deployment study rather than a controlled experiment. The findings are reported as quantitative operational outcomes of a production DTI implementation. Controlled replication across multiple institutions is required before generalizing these findings.

imaware is a direct-to-consumer laboratory diagnostics company offering at-home collection for panels including cardiometabolic, thyroid, hormonal, and inflammatory markers. Prior to DTI implementation, imaware managed approximately 105,000 diagnostic records across fragmented data systems with no unified quality signal.

6.0 Study Design

Setting: Single-organization retrospective before/after comparison. **Data:** 105,000 diagnostic records across imaware’s laboratory information system. **Intervention:** Full DTI implementation via SuperTruth IntegrityNet integration. **Comparison period:** 6-month pre-implementation baseline vs. 6-month post-implementation observation. **Primary outcomes:** (1) data preparation cycle time; (2) analyst hours allocated to data quality tasks; (3) identifiable customer segment structure. **Limitations:** No randomization, no control arm, single organization. Confounders including organizational learning effects and concurrent infrastructure improvements cannot be excluded.

6.1 Implementation

SuperTruth integrated with imaware’s laboratory information system (LIS) via structured API integration with the CLIA-certified processing laboratory. DTI scoring was applied to all records at ingestion. The integration required no data movement: IntegrityNet’s zero-copy connectors maintained data in imaware’s existing systems while appending DTI metadata.

6.2 Results

Outcome 1: Data preparation cycle time.

Metric	Pre-DTI	Post-DTI	Change
Data preparation cycle time	~3 weeks	~2 hours	–95%
Analyst hours/month on data quality	~200 hours	~0 hours	–200 hrs/month

Prior to DTI, preparing a data set for analysis required approximately three weeks of manual review, deduplication, and quality assessment. Post-DTI, the same preparation cycle completed in approximately two hours using DTI score thresholds to filter records programmatically, a 95% reduction. Approximately 200 analyst hours per month previously allocated to data hygiene tasks were freed for higher-value work.

Outcome 2: Latent segment identification.

DTI-based segmentation of the 105,000-record dataset identified a customer segment that had been indistinguishable in the pre-DTI fragmented data view. This segment accounted for 20% of imaware’s revenue. The segment was characterized by consistent concordance between at-home and clinical laboratory results, high Recency scores (frequent re-testing behavior), and multi-panel Breadth. These properties were measurable only after DTI scoring unified and ranked the underlying records into a queryable structure.

Qualitative assessment. Brodie Flanders, CEO of imaware, stated: “The lab industry has never had a trust standard. DTI created one.”

6.3 Prospective Validation Protocol

The following validation study design is proposed to assess DTI’s impact on downstream model performance across independent institutions. This study has not yet been conducted; the protocol is published here to establish a pre-registered methodological framework.

Study objective: To determine whether training clinical AI models on DTI-filtered data (records above a specified score threshold) produces measurably better performance than training on unfiltered data from the same source population.

Data sources: Three independent health systems contributing de-identified records under IRB approval, targeting a minimum of 50,000 records per site across at least two modalities (laboratory + wearable or laboratory + clinical notes).

Experimental conditions: For each site, three model training conditions: (1) unfiltered baseline (all records), (2) DTI threshold ≥ 70 (Silver-tier filter), (3) DTI threshold ≥ 80 (Gold-tier filter).

Outcome metrics: Model calibration (Brier score), discrimination (AUROC), and positive predictive value on two clinically validated prediction tasks: 30-day readmission and HbA1c trajectory.

Hypothesis: Models trained on DTI-filtered data will show statistically significant improvement in Brier score relative to unfiltered baseline, with the largest improvement in the Gold-tier condition. We further hypothesize that improvement will be greater for sites with higher pre-DTI data fragmentation.

Planned analysis: Mixed-effects model with site as a random effect. Significance threshold $p < 0.05$ with Bonferroni correction for multiple comparisons. Pre-registration on OSF prior to data collection.

6.4 Limitations

The imaware implementation covers laboratory diagnostics data only; breadth scores for this data set are accordingly lower than for multi-modal datasets. The concordance dimension benefited from imaware’s dual-source architecture (at-home collection followed by clinical confirmation), which is not universal across laboratory operators.

The reported time reduction from three weeks to two hours reflects a before/after comparison within a single organization, not a controlled experiment with a held-out condition. Confounding factors — including organizational learning effects and concurrent data infrastructure improvements — cannot be fully excluded. The figure should be treated as an operational benchmark, not a causal estimate.

The revenue intelligence finding (20% of revenue from a previously invisible segment) was surfaced post-hoc through DTI-enabled segmentation. Prospective replication studies are required to establish whether similar latent segments exist across other diagnostics operators and whether DTI-based segmentation reliably surfaces them.

7. Discussion

7.1 Regulatory Context

The 21st Century Cures Act [17] mandated open APIs for health data access and defined information blocking prohibitions, accelerating the volume of health data available to AI systems without establishing quality standards for that data. TEFCA [18], implemented by the Office of the National Coordinator for Health IT, defines a trust framework for data exchange organizations (QHINs) but addresses organizational trustworthiness rather than record-level data quality.

FDA's 2021 AI/ML SaMD action plan [3] and subsequent draft guidance on predetermined change control plans identify training data quality as a critical gap in the AI medical device framework. The guidance states that developers must document "the characteristics of the training data" but does not specify how those characteristics should be quantified. DTI provides one operationalization: a standardized, auditable score that can be included in SaMD submissions to document the quality of data used for training and validation.

7.2 Implications for Federated Learning

In federated learning architectures where model weights are aggregated across institutions, DTI scores can serve as a data quality weight in the aggregation step. Institutions contributing records with high average DTI scores should carry greater weight in gradient aggregation than institutions contributing lower-quality data. This is analogous to sample weighting in classical meta-analysis, where study quality weights are standard practice.

7.3 Consent Portability

As ConsentOS implements cryptographic consent attestation, DTI scores can function as consent-aware trust signals in cross-institutional data sharing under TEFCA. A record with a Platinum composite DTI score demonstrates high confidence across all eight dimensions simultaneously. Receiving institutions can inspect the per-dimension breakdown — including the Consent sub-score specifically — to make data use decisions based on auditable, structured quality evidence rather than re-adjudicating raw consent documentation from the source institution. For regulatory submission contexts, deployment configurations may optionally enforce a minimum Consent sub-score floor (e.g., $\text{Consent} \geq 80$) as a prerequisite for Platinum classification, providing a hard guarantee on consent quality independent of the composite weighting.

7.4 Limitations and Future Work

DTI's default weights reflect current regulatory guidance and clinical expert consensus and will require revision as the regulatory landscape evolves. The validation dimension relies on existing peer-reviewed literature, which has known limitations including publication bias and limited coverage of emerging biomarkers. The framework currently supports laboratory, wearable, clinical, and SDOH data modalities; genomics and imaging data require modality-specific extensions that are in development.

Future work includes: (1) execution of the prospective multi-site validation protocol described in Section 6.3, comparing DTI-filtered model performance against unfiltered baselines on readmission prediction and HbA1c trajectory tasks; (2) controlled sensitivity analysis studies examining how composite score orderings respond to weight perturbation across multiple institutional deployments; (3) extension of the concordance methodology to imaging and genomic modalities; and (4) international regulatory alignment covering CE marking under the EU MDR and GDPR consent tier mapping.

8. Conclusion

Health AI systems operate on data of widely varying quality, provenance, and consent coverage. Treating this data as uniformly trustworthy produces models whose failure modes are difficult to detect and harder to attribute. The Data Trust Index provides a record-level integrity signal that is composable, interpretable, and actionable across regulatory contexts from exploratory research to FDA submission.

The imaware case study demonstrates that DTI-based data governance can simultaneously reduce operational costs, surface hidden commercial value, and produce auditable data quality documentation. As AI-based clinical decision support moves from research to deployment, the absence of standardized data quality metrics represents a systemic risk. DTI addresses that risk by making data trust a first-class property of every record in the pipeline.

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Conflict of Interest

The author is Co-Founder and Chief AI Officer of SuperTruth Inc., the company that develops and commercializes the Data Trust Index framework described in this paper. The imaware case study was conducted as part of a commercial engagement between

Data Availability

The imaware deployment dataset used in Section 6 contains protected health information and is not publicly available. Aggregate metrics are reported with organizational permission from imaware. The DTI scoring algorithm and pipeline architecture are described in sufficient detail in Sections 3-5 to permit independent implementation. A reference corpus of public knowledge about the DTI framework is available at <https://supertruth.ai/api/export/corpus> under a CC-BY-4.0 license.

About the Author

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